

# Is the volume-law horizon a postulate or a consequence? A reduction to a driven non-equilibrium conjecture for Structural Entropy Dark Energy

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## Abstract

Structural Entropy Dark Energy (SEDE) reproduces the cosmological data with no fitted dark-sector parameter, at the price of one foundational input: the cosmic apparent horizon carries volume-law entropy (Barrow exponent  $\Delta = 1$ ) — scaling with the enclosed volume rather than the boundary area. We ask how much of that input is irreducible. The postulate decomposes into *state, form, scale, and count*; state, form and scale are first-principles or reducible given the standard identification of the gravitational horizon entropy with an entanglement entropy — with the caveat, stated up front, that a *volume-law* coefficient (unlike the area law’s universal  $A/4G$ ) is state-dependent, so *scale* reduces only *given* the Bousso-saturated horizon state that fixes  $s_{\text{grav}}$ : a conditional, not a universal coefficient. That leaves the bulk-versus-boundary count as the residue — which a Bekenstein no-go shows cannot follow from any energy bound. We then analyse the *selection* of the volume branch dynamically, as a motivated conditional rather than a derivation: because gravity’s long-range cooperative coupling clears the ordering threshold only with a size-dependent ratio, the two-phase free energy did not exist at the horizon’s birth; grown continuously from Planckian size, the horizon passes *through* the ordering bifurcation with Barrow entropy tilting the nascent well toward volume, so  $\Delta = 1$  is *favoured* at birth, and long-range hysteresis locks each horizon to its birth branch — the cosmic horizon to volume, an abruptly-formed black hole to area. We are explicit about the limits of this argument: the tie-breaking tilt is itself the Barrow- $\Delta = 1$  free energy, the two-well landscape presupposes an accessible volume phase, the mean-field treatment is extrapolated to the  $N \sim O(1)$  quantum-gravity regime where its tools are not controlled, and the one concrete model of de Sitter holography (double-scaled SYK) currently maps to the *area* branch — so the mechanism relocates and narrows the postulate, it does not eliminate it. Its lasting value is to convert an untestable counting postulate into a falsifiable one: a pre-registered point prediction  $\Delta = 1$ , tested to a forecast  $\sigma(\Delta) \approx 0.09$  by DESI DR3 + Euclid. On public data the contrast is already sharp — in a *profile*-likelihood analysis (the marginalised posterior awaits DR3+Euclid) the structure-gated family *prefers* the volume endpoint ( $\Delta = 0.93$  [0.83, 1.02]) that the un-gated Barrow family pays  $\Delta\chi^2 \approx 2400$  to reach. The marginalised model-level statement is more modest: at equal parameter count the zero-parameter  $\Delta = 1$  model is favoured over  $\Lambda$ CDM at the cosmology paper’s  $\Delta\text{DIC} \approx -3$  (a point-fit gives  $\Delta\chi^2 \approx 2.9$ , but the DIC is the honest headline). The cosmological results of the cosmology paper stand independently of the reduction attempted here.

**Keywords:** horizon thermodynamics; generalised entropy; Barrow entropy; holographic dark energy; de Sitter holography; entanglement entropy; non-equilibrium statistical mechanics; entropic gravity; dark energy

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## 1 Introduction

Horizon thermodynamics is the most robust pointer we have toward the microscopic structure of gravity. The first law on a causal horizon yields the Einstein equations [1], the apparent-horizon first law yields the Friedmann equations [2], and the Bekenstein–Hawking area law  $S = A/4$  organises black-hole and cosmological thermodynamics alike. Yet the area law is an assumption about a *state*: it is the entanglement entropy of a particular (ground-like) horizon state, and generalisations — Tsallis–Cirto non-extensive entropy, Barrow’s fractal-horizon  $S \propto A^{1+\Delta/2}$  [3], and the holographic dark-energy programme built on them [4] — explore what happens when that assumption is relaxed.

Structural Entropy Dark Energy (SEDE; the cosmology paper [5]) is one such model, distinguished by sourcing dark energy from horizon entropy *gated by structure growth*,

$\rho_{\text{DE}} = T_{\text{AH}} s_{\text{grav}} f_{\text{sat}}(z)$ . Following the cosmology paper, this is *holographic* dark energy —  $\rho_{\text{DE}}$  is added as a single fluid to an otherwise standard-GR Friedmann equation, not a Barrow-*modified*-gravity cosmology in which the horizon entropy reshapes the background; the volume-law entropy sources  $\rho_{\text{DE}}$  alone (so the modified-gravity Barrow/BBN bounds do not apply, and the postulate examined below is one of *state*, not of dynamics). Its appeal is that the entire dark sector adds no *fitted* parameter once  $\Omega_{\text{m}}$  and the one postulate are fixed: the H-coupling  $\lambda = 1 - \Delta/2 = \frac{1}{2}$ , the equation of state, the structure coupling  $\gamma$ , and the sound speed all follow. The one input is the **volume-law postulate**: the horizon entropy is the volume-law ( $\Delta = 1$ ) entanglement entropy, not the area law. The cosmology paper states plainly that this is the model’s single irreducible assumption, motivated but not derived, and that it is falsifiable —  $\Delta$  is measured by upcoming surveys.

This paper asks the prior question: *how irreducible is it, really?* We are not trying to solve quantum gravity; we are trying to determine the minimal, sharply-posed statement that the cosmology actually rests on, and to reduce everything reducible to it. The result is a reformulation — one that ends with a shorter, sharper list of residuals than the postulate it replaces. We trade a static, untestable counting postulate for a physical picture in which  $\Delta = 1$  is selected *at the horizon’s birth*, carried through the ordering bifurcation that gravity’s N-dependent coupling ( $J/J_c = 2\pi G m^2 N$ ) forces at Planckian size, with Barrow’s entropic tilt breaking the tie and exact long-range hysteresis locking every horizon to its birth branch. The remaining hand-off to de Sitter holography is sharpened to a single algebraic question — the type,  $\text{II}_1$  or  $\text{II}_\infty$ , of the FRW static-patch observer algebra. Several steps borrow established machinery: the Cohen–Kaplan–Nelson holographic energy bound [6] for the scale; the Page curve and eigenstate thermalisation for the state; the Campa–Dauxois–Ruffo classification of long-range systems [7] for the connectivity; Ryu–Takayanagi min-cut for the count $\leftrightarrow$ connectivity link; and the double-scaled Sachdev–Ye–Kitaev (SYK)/de Sitter correspondence [8,9] for the closure attempt. The contribution is the chain, and the precise localisation of what remains open.

Throughout, every quantitative claim is a runnable experiment that the accompanying code runs with validation assertions.

## 1.1 Relation to prior work

SEDE sits at the confluence of several programmes, and its closest relative deserves explicit positioning.

Entropy-production cosmic acceleration (GREA). The General Relativistic Entropic Acceleration theory of García-Bellido and Espinosa-Portalés [10,11] is the nearest existing model: it too abandons  $\Lambda$ , sources the late-time acceleration from cosmic-*horizon* entropy, takes the Helmholtz free energy  $F = U - TS$  (not matter alone) as the gravitational source, and makes the breaking of time-reversal by *entropy production* the engine of acceleration. We regard GREA as prior art for “acceleration from horizon entropy production without  $\Lambda$ ”, and our driven-NESS picture (§4–6) is conceptually continuous with it. The differences are sharp and testable: (i) GREA uses the boundary (area) horizon entropy and a single parameter  $\alpha$  (the horizon-to-curvature ratio), whereas SEDE’s defining claim is the volume entropy ( $\Delta = 1$ ) — precisely the counting residue this paper isolates; (ii) GREA’s entropy growth is driven by the horizon’s own *expansion*, whereas SEDE’s is *gated by structure formation* (the  $f_{\text{sat}}$  factor), which locks  $w(z)$  to the growth history and yields a phantom crossing rather than GREA’s quintessence-like  $w(a)$ ; (iii) SEDE’s dark sector adds no *fitted* parameter. Notably, were SEDE’s residue to resolve to the *area* branch ( $\Delta = 0$ ), its phenomenology would collapse toward GREA’s — so the  $\Delta$  measurement (§7) that decides SEDE’s residue also discriminates the two models. A holographic reading of GREA has recently appeared [12], and a broader open-system view of gravitational non-

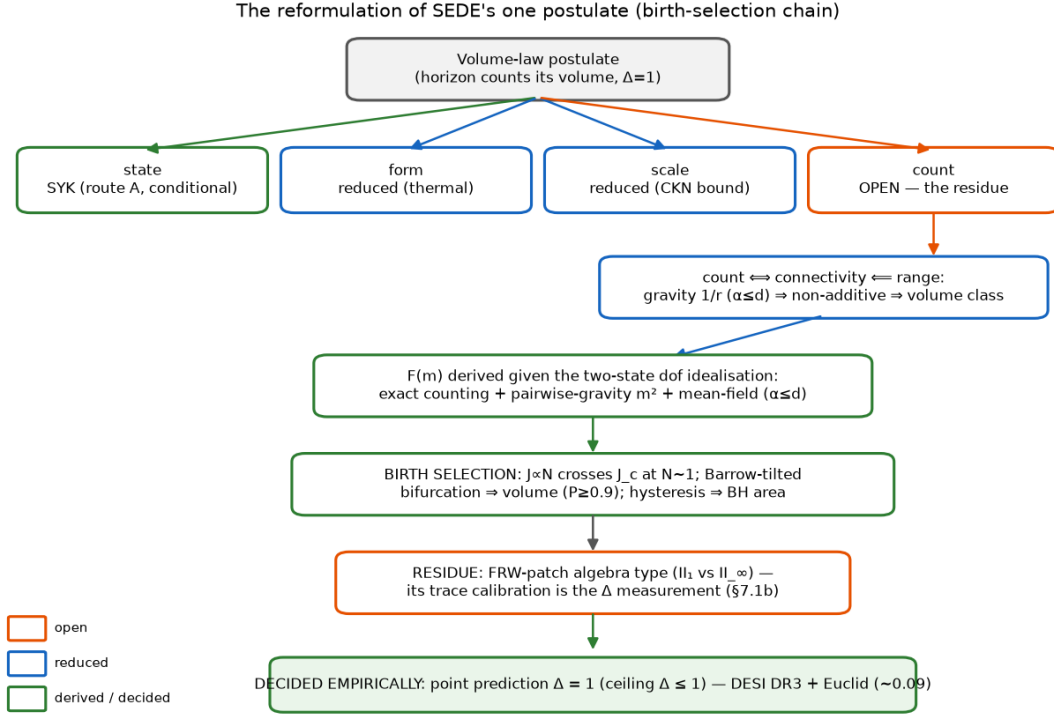


Figure 1: The reduction in outline. The volume-law postulate splits into *state*, *form*, *scale* and *count*; the first three are derived or reduced (green/blue) and the count is the residue (orange). The count reduces through connectivity and interaction range to a birth-branch selection: the  $N$ -dependent coupling carries the grown horizon through the ordering bifurcation onto the Barrow-tilted (volume) branch, where long-range hysteresis locks it (a *state-selection* hysteresis of the entropy branch — distinct from the unrelated “cosmological hysteresis” of cyclic cosmologies, which refers to the  $\oint p dV$  loop), leaving a single dS-holography question that the  $\Delta$  measurement decides empirically. Each box names the experiment that establishes it.

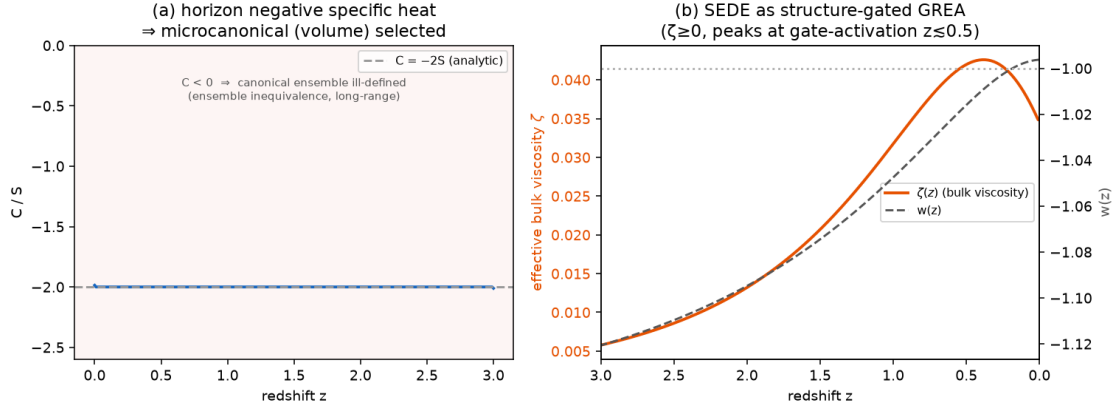


Figure 2: Two consequences drawn from the new literature. (a) The de Sitter horizon has negative specific heat ( $C = -2S$ ); with gravity strongly long-range this makes the ensembles inequivalent and the canonical (area-law) accounting ill-defined, selecting the microcanonical (volume) branch (§4). (b) SEDE’s structure-driven entropy production maps to a positive (second-law) effective bulk viscosity  $\zeta(z)$  that peaks at the gate-activation epoch ( $z \sim 0.5$ ; it tracks  $d \ln f_{\text{sat}}/d \ln a$ , not the  $z \approx 2$  star-formation peak), casting SEDE as a structure-gated member of the GREA/entropic-dark-energy family;  $w(z)$  crosses  $-1$  at the viscous–dilution balance.

equilibrium thermodynamics is developing [13,14]; SEDE’s contribution to that line is the reduction of the volume-law *counting* input to a driven steady state. The connection is also quantitative: GREA’s entropy production maps to an effective bulk viscosity with negative pressure, and SEDE’s structure-driven entropy production  $d \ln f_{\text{sat}}/d \ln a > 0$  maps to a *positive* (second-law-respecting) effective bulk viscosity  $\zeta(z) = \rho_{\text{DE}} (d \ln f_{\text{sat}}/d \ln a)/(9H)$  that peaks at the structure-formation epoch rather than tracking the horizon expansion; the  $w = -1$  crossing is the GREA negative-pressure balance, structure-timed. SEDE is, in this sense, a *structure-gated* member of the GREA/entropic-dark-energy family — the gate being exactly what makes its  $w(z)$  lock to the growth history (the §5 phase-lock) rather than to the horizon’s own size.

Holographic dark energy and generalised entropy. SEDE is, at the background level, a structure-gated Barrow holographic dark energy in the lineage founded by Li [15]: the CKN bound [6] fixes the scale, Barrow’s fractal-horizon entropy [3] and its holographic-DE use [4] fix the form (the  $\lambda = 1 - \Delta/2$  relation is Saridakis’s; our increment is the  $\Delta = 1$  selection, not that relation), and the Tsallis–Cirto non-extensive entropy [16] supplies the  $\delta \leftrightarrow \Delta$  map. The CKN-vs-area-vs-volume tension these inputs create is mapped by Manoharan [17]; our addition is the *derivation* question — which of these inputs is irreducible — and the answer that only the count is.

Horizon thermodynamics and entanglement. The first-law derivations of Einstein [1] and Friedmann [2] equations underlie the construction; the entanglement-equilibrium refinement [18] is the basis of route B; the Ryu–Takayanagi min-cut [19] supplies the count  $\Leftrightarrow$  connectivity link; and the volume-law = thermal identification rests on eigenstate thermalisation [20–22].

**de Sitter holography.** The residue — “ $\dim \mathcal{H}(\text{static patch}) = e^{\text{Area}}$  or  $e^{\text{Vol}}$ ?” — is posed in the language of de Sitter holography [23,24] and is the target of the double-scaled-SYK/de Sitter correspondence [8,9,25], which we use in the closure attempt (§7). The maximal-scrambler state invokes the chaos bound [26] and SYK random-matrix uni-

versality [27].

Many-body and complexity inputs. The long-range/non-additivity classification is [7], and the  $\alpha$ -controlled volume $\leftrightarrow$ area (and intermediate/fractal) crossover of long-range entanglement entropy is already characterized in the literature [28–31] — the known crossover we apply to a horizon count (§ the count companion), not a new phenomenon. The prototype’s free-fermion *negative* (volume-law needs interactions, not free transport) is consistent with the monitored free-fermion entanglement literature [32], and that genuine volume-law non-equilibrium steady states exist is established in driven many-body systems [33] — a condensed-matter precedent for, not a derivation of, the driven-NESS we invoke.

Structure-sourced dark energy. The general idea that dark energy is sourced by cosmic structure predates SEDE in several distinct mechanisms: Gough’s information dark energy ( $\approx 2008$ ); the spacetime-fragmentation/“FRW-islands” proposal of Hossain [34]; and the cosmological-backreaction and averaging programme (Buchert; Wiltshire’s timescape). SEDE’s cosmology paper credits and distinguishes these. What is specific to SEDE — and, to our knowledge, without precedent — is the *combination* the present paper analyses: the source is the volume-law horizon entropy ( $\Delta = 1$ , a counting claim), it is driven by structure through the  $f_{\text{sat}}$  gate, and the dark sector adds no *fitted* parameter. The closest entropic relative remains GREA (above), which differs in using the boundary (area) entropy. A systematic priority search returned no work combining these three elements; this paper concerns that horizon-entropy foundation. The most recent independent convergence is Pandey [35], who obtains late-time acceleration and suppressed growth from the *configuration entropy* of the collapsing matter field as a backreaction within GR — reaching SEDE’s qualitative conclusion by a different route (matter-sector entropy, not the horizon; no derived w-crossing; no  $\Delta$ ), which we welcome as corroboration of the direction. A parallel generalised-mass-to-horizon-entropy programme [36, 37] links a modified horizon entropy to growth via a *modified-Friedmann* route, and so inherits the BBN  $\Delta$ -bounds that SEDE’s holographic-fluid scope evades.

## 2 The postulate, decomposed

“Volume-law” conflates four claims of very different status:

- state — the horizon degrees of freedom are maximally entangled (thermal), not in a low-entanglement ground state;
- form — the entropy scales as the volume,  $S \propto V$ ;
- scale — the magnitude is  $\rho_{\text{DE}} \sim \rho_{\text{crit}}$ , not  $\rho_{\text{Planck}}$ ;
- count — the *number* of horizon dof grows as the bulk ( $N \propto R^{d-1}$ ) rather than the boundary ( $N \propto R^{d-2}$ ).

Since  $S \sim (\text{state factor}) \times N$  and the Barrow deformation enters as  $S \propto A^{1+\Delta/2} \propto R^{(d-2)(1+\Delta/2)}$ , the exponent  $\Delta$  is fixed by the count (area  $\rightarrow \Delta = 0$ , volume  $\rightarrow \Delta = 1$  in  $d=4$ ); the state only sets whether the prefactor is maximal. So the postulate’s empirical content —  $\Delta$  — is the counting claim, and the other three are separable. The rest of the paper derives or reduces state, form and scale, and isolates count.



### 3 How derivable is the postulate? The route map

The horizon-entropy identification (assumed, not discharged). Routes A/B below characterise the *matter-field subsystem entanglement entropy*  $S_{\text{ent}}$  (Page/ETH/SYK) — UV-divergent, species-dependent. The dark sector is sourced by the *gravitational horizon entropy*  $S_{\text{grav}} = A/4G$  (Cai–Kim/Jacobson) — UV-finite, geometric.  $S_{\text{ent}} \equiv S_{\text{grav}}$  is a long-standing open problem in quantum gravity ([38, 39]), established only for the Rindler vacuum. We adopt it (the induced-gravity premise) and flag every dependent step as conditional. In particular Route A fixes the entropy *prefactor* (maximal scrambling), which is orthogonal to the bulk-vs-boundary *exponent* that is the count; and Route B’s “volume-law is generic” is a statement about  $S_{\text{ent}}$ , not  $S_{\text{grav}}$ . Accordingly the *state*, *form*, and residue-to-range steps are argued only conditionally on  $S_{\text{ent}} = S_{\text{grav}}$  — not established outright.

Five routes (`run_deriv_{A..E}*.py`):

A — de Sitter holography / maximal scrambler. A Majorana-SYK diagonalisation confirms the horizon *state* is maximally entangled / volume-law: chaotic level statistics ( $\langle r \rangle \approx 0.71$ ) and Page-value saturation [40] ( $S/S_{\text{Page}} \approx 0.99$ ). But SYK is all-to-all / geometry-free and cannot pose the bulk-vs-boundary count; the count is the genuine open dS-holography sub-target.

B — entanglement first law in a thermal state. Jacobson’s derivation gets the area-law Einstein equations from maximal entanglement of the *vacuum*. Redone around a *thermal* (de Sitter Gibbs) reference, the entanglement entropy acquires a volume term that dominates the area term once the region exceeds the thermal length  $\lambda_{\text{th}} = 1/T$ . Volume-law is thus the generic thermalised behaviour, not exotic. At the de Sitter temperature alone the horizon sits just on the area side ( $R_{\text{H}}/\lambda_{\text{th}} = 1/2\pi$ ); volume-law dominance requires thermalisation above the de Sitter bath (toward Planck = maximal scrambling), whereupon the density is Planckian — the CKN scale. So the *form* reduces to thermalisation, leaving the *scale* to CKN.

C — Verlinde emergent gravity. A volume-law de Sitter entanglement entropy is exactly what Verlinde’s emergent-gravity programme invokes for apparent dark matter; one volume-law scale gives both  $\rho_{\text{DE}} \sim \rho_{\text{crit}}$  and  $a_0 = cH_0/2\pi$  ( $\approx 0.87 \times$  the radial-acceleration-relation value). A unification motivation that ties the postulate to an independent anomaly; it inherits that programme’s open issues.

D — gravitational non-additivity. The Tsallis–Cirto map  $\delta = 1 + \Delta/2$  makes  $\Delta = 1 \iff \delta = 3/2$  exact, but pinning  $\delta$  from a measurable non-additivity fails: real cluster kinematics are Gaussian ( $q \approx 1.01$ ), not  $q \approx 1.5$  — the weakest route.

E — no-go from energy bounds. The Bekenstein bound on the horizon energy gives  $S \leq 2\pi R E/\hbar c = \frac{1}{4}(A/\ell_{\text{P}}^2)$  — *exactly* the area law (verified  $S_{\text{Bek}}/S_{\text{area}} = 1.000$ , the known  $10^{122}$ ). The volume law therefore cannot come from any maximum-entropy/energy argument; it exceeds the bound by precisely  $R/\ell_{\text{P}} \sim 10^{61}$  (the “seam”). This proves the missing ingredient is a state/counting input and fixes its size.

*Net:* state (A) and form (B) are argued (conditional on  $S_{\text{ent}} = S_{\text{grav}}$  — and “state” fixes the entropy *prefactor*, not the count *exponent*), the scale (CKN) is consistent with the data, the canonical area-law horn is genuinely removed by ensemble inequivalence (§4); the irreducible residue is the bulk-vs-boundary count.

## 4 Reducing the residue to a driven NESS

The count is not a free coin-flip — but here we must be careful to avoid a circularity, because the entanglement class cannot simply be *posited*. It is fixed by the Hamiltonian and the state, and the decisive fact is that the entanglement area law is a theorem with a hypothesis: it is proved for *local* (finite-range) Hamiltonians — Hastings [41] rigorously in 1D via Lieb–Robinson bounds, Brandão–Horodecki [42] from exponential decay of correlations, reviewed in Eisert–Cramer–Plenio [43] — and the proof *requires* exponential clustering. Gravity is  $1/r$  ( $\alpha = 1$ ), strongly long-range ( $\alpha \leq d$ ) and non-additive [7], and there the hypothesis fails: in an explicit gapped free-fermion model we find the correlation function decays as a *power law* for  $\alpha = 1$  (slope  $-1.5$ ) versus *exponentially* for the short-range reference (slope  $-6.9$ ), exactly as Koffel–Lewenstein–Tagliacozzo [44] found. So the area-law theorem does not apply to a gravitationally-bound horizon: area-law is *not guaranteed*, and the count is genuinely open between area and volume with neither the default. We are explicit about what this does and does not give. It does *not* by itself derive volume — a noncritical long-range ground state can still be area-law [45], and indeed the gapped ground-state block entropy stays bounded for every  $\alpha$  we test. The volume law additionally requires the horizon *state* to be thermal / maximally scrambled (route A), where entanglement is volume-law; and even that fixes the *state*, not the *count*. The super-extensive non-additivity ( $u \propto N^{1-\alpha/d}$ , slopes 0.53 in  $d=2$ , 0.68 in  $d=3$  vs 0.50, 0.67) is the thermodynamic face of the same long-range property. The conclusion is that volume is the *motivated, non-default* class for a self-gravitating thermal horizon — area-law arises only when the holographic bound intervenes for an *isolated, equilibrium* horizon (a black hole saturates the Bekenstein bound, route E, and relaxes to area-law) — while the bulk-vs-boundary count itself remains the open dS-holography residue. One caveat qualifies this: the area-law theorems are statements about many-body systems with a specified Hilbert space and Hamiltonian, and what the horizon’s degrees of freedom *are* — and what Hamiltonian governs them — is precisely the open dS-holography question. So this is a motivated *conditional* — *if* the horizon is a long-range many-body system, *then* area-law is not forced and volume is favoured — not a theorem about the actual horizon. What it establishes is that the entanglement class is read off the (modelled) Hamiltonian rather than assumed — which is what defeats the circularity charge — not that the count is derived.

This same long-range property has a second, independent consequence that bears on the canonical-vs-microcanonical  $\lambda$  ambiguity (whether  $\rho_{\text{DE}} \propto H^2 f_{\text{sat}}$ , area, or  $\propto H f_{\text{sat}}$ , volume). A non-additive system has *inequivalent* statistical ensembles [7], the signature being negative specific heat — and the de Sitter horizon indeed has  $C = -2S < 0$  throughout its history. For such a system the canonical ensemble is ill-defined; the isolated self-gravitating horizon must be treated *microcanonically* (by its state count). The canonical accounting is precisely the area-law ( $\lambda = 1$ ) branch, so ensemble inequivalence removes it as unphysical and selects the microcanonical (volume,  $\lambda = 1/2$ ) branch the model uses. This does not close the residue — whether the microcanonical state count is itself volume is the dS-holography question — but it eliminates one of its two horns from first principles. The cosmic horizon differs in one respect: it is continuously *driven* by structure formation (the  $f_{\text{sat}}$  gate,  $dS_{\text{struct}}/dt > 0$ ; present drive  $\approx 0.45$ ). This turns the black-hole/cosmic asymmetry into a discriminator — equilibrium  $\rightarrow$  area, driven NESS  $\rightarrow$  volume — and reduces the residue to a single, falsifiable statement:

**Driven-NESS conjecture.** A strongly-long-range horizon, driven by structure formation, settles in the volume-law steady state rather than relaxing to the area-law equilibrium.



Two sections sharpen this conjecture into calculations. The *selection* half is replaced by a mechanism that needs no steady-state argument at all — birth at the ordering bifurcation, §5 (iii) — while the *driving* half is localised in a one-line activation kinetics whose solution is exactly the cosmology paper’s gate (§7.1) [5]: the conjecture survives as the statement that structure deposits *populate* the volume capacity, not that they select the branch.

## 5 Selection of the volume branch: birth at the ordering bifurcation

We ask what selects the volume branch dynamically, separating what is robust from what is conditional. Four results, each backed by an explicit calculation.

- (i) Barrow entropy supplies a driving slope, not a landscape. The bare Barrow free energy  $F(\Delta) = E - T_{\text{AH}} S_{\Delta}$ , with  $S_{\Delta} = (A/A_0)^{1+\Delta/2}$  and the Misner–Sharp energy, is *monotonic* in  $\Delta$  —  $F(0) = 0$  by the first law, and  $F$  decreases to  $\Delta = 1$  with no barrier and no second minimum. Bistability is therefore not a consequence of Barrow entropy: Barrow supplies the entropic drive toward  $\Delta = 1$ , not the two-phase structure. (This is the actual content of the GSL argument of §8.1 — it drives  $\Delta \rightarrow 1$ , it does not by itself produce a spinodal.)
- (ii) Gravity supplies the barrier — deep in the ordered phase. A two-phase (area $\leftrightarrow$ volume) free energy is bistable only if the horizon degrees of freedom have a cooperative coupling above the ordering threshold,  $J \geq J_c$ . That coupling is the largest eigenvalue of the gravitational coupling matrix,  $J = \lambda_{\text{max}}(W_{\text{grav}})$ ,  $W_{ij} = g/r_{ij}^{\alpha}$  — the *same* super-extensive quantity behind the non-additivity of §4 ( $\lambda_{\text{max}}/\text{row-sum} = 1.02$ ), so volume-counting and volume-locking are one number. Comparing it to  $J_c = T_{\text{AH}}$  in the same units, for  $N \sim A/4$  Planckian dof on the horizon 2-sphere, gives  $J/J_c = 2\pi G m^2 N \sim 2\pi N \sim 10^{123}$  (the non-additive scaling  $\lambda_{\text{max}} \propto N^{1-\alpha/d} = N^{1/2}$  is confirmed numerically — slopes 0.52/0.67, a short-range  $\alpha > d$  control bounded at  $N^{0.01}$ ). Gravity clears the threshold by of order the holographic count: the two-phase well exists, its minima are the pure-area and pure-volume phases, and the barrier is  $\sim N \cdot T_{\text{AH}}$ . This places the horizon deep in the ordered phase, not near a spinodal — *today*. But the same formula contains the key to the selection:  $J/J_c = 2\pi G m^2 N$  is  $N$ -dependent, so the barrier did not exist at the horizon’s birth. The Landau form is not otherwise assumed *given the two-state dof idealisation* — the one place the  $S_{\text{ent}} = S_{\text{grav}}$  tier re-enters, sourced micro-structurally below, and the qualification that governs everything in this item: within that idealisation all three terms of  $F(m)/NT = -(J/2)m^2 + \theta m + [m \ln m + (1-m)\ln(1-m)]$  are derived — the mixing entropy is exact combinatorics of choosing which dof participate; the  $m^2$  term is the pairwise  $1/r$  binding of the participating dof (measured exponent 2.008), with  $J = \lambda_{\text{max}}(W_{\text{grav}})$  the same number as above; and mean-field thermodynamics is *exact* for  $\alpha < d$  [7] (verified: a Kac-normalised  $\alpha = 1$  long-range Ising sits at the mean-field  $T_c$  while the nearest-neighbour control sits at Onsager’s, far below). Even the two-state (boundary-/bulk-entangling) character of the dof is not an idealisation inserted by hand: in the random-tensor-network model of holographic states [46] the entanglement computation is *exactly* dual to an Ising model — each tensor carries a binary (identity/swap) replica variable whose domain wall is the Ryu–Takayanagi min-cut — and we verify both the min-cut law (a three-leg region behind a bulk

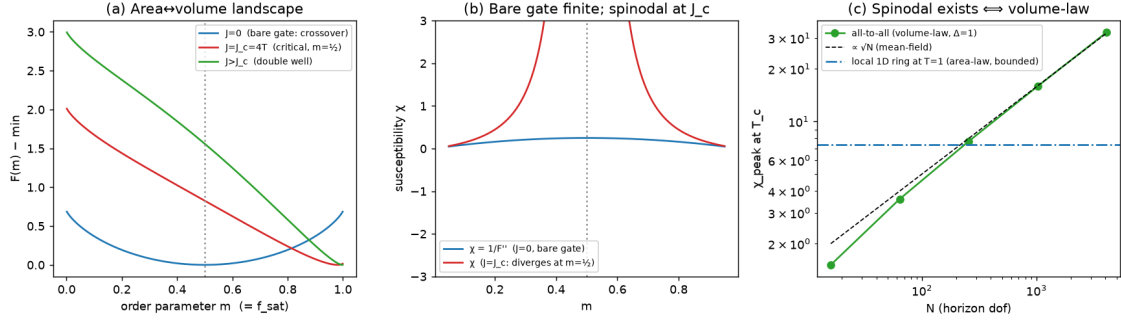


Figure 3: The two-phase horizon free energy. (a) Smooth crossover for the bare gate ( $J = 0$ ); a spinodal develops only for  $J \geq J_c$ . (b) The bare-gate susceptibility is finite; it diverges at the spinodal. (c) The cooperative coupling  $J = \lambda_{\max}(W_{\text{grav}})$  is the gravitational binding itself: all-to-all (volume-law) connectivity gives  $\chi_{\text{peak}} \propto \sqrt{N}$  (so  $J/J_c \sim N$ , far above threshold), while local (area-law) connectivity is bounded. Gravity thus makes the well bistable — but *deeply* so today, which is why selection happens at the birth bifurcation (result iii, Fig. 4) rather than by barrier crossing (the horizon’s membrane hydrodynamics, result iv, is a separate matter).

bottleneck saturates at  $1 \cdot \ln \chi$ , not  $3 \cdot \ln \chi$ ) and the duality itself (the two-state partition sum  $Z_A/Z_0$  reproduces the measured purity to  $< 1\%$  at bond dimension  $\chi \geq 4$ ). The order parameter thereby acquires a microscopic definition —  $m$  = the fraction of horizon bonds on the bulk side of the min-cut — and the residual assumption shrinks to the tensor-network description of the horizon state itself: the  $S_{\text{ent}} = S_{\text{grav}}$  tier restated micro-structurally, one assumption where two stood before.

- (iii) Selection at the ordering bifurcation, and exact hysteresis. A barrier  $\sim N \cdot T_{\text{AH}}$  is *uncrossable*: Kramers escape is suppressed by  $e^{-N \cdot \Delta f}$ ,  $N \sim 10^{122}$  — so “relaxation to the global minimum” cannot be the selection mechanism, and we do not invoke it. Instead, the  $N$ -dependence of (ii) supplies the mechanism. The cosmic apparent horizon grew continuously from  $N \sim O(1)$ , so it passed *through* the ordering bifurcation as the double well formed around it: near threshold the barrier vanishes quadratically,  $\Delta f \propto (J - J_c)^2$  (measured exponent 1.89), while the Barrow tilt of (i) is already present — and a *tilted* pitchfork has no branch ambiguity. The minimum continuously connected to the disordered  $N \sim 1$  state is the tilt-favoured (volume) branch; the area minimum is born *metastable* only at the spinodal  $J_{\text{sp}} > J_c$  ( $J_{\text{sp}} \approx 5.6$  at tilt  $b = 0.3$ ) and is never occupied. The swept trajectory confirms it: gradient flow tracks continuously into the volume well (final  $m = 1.00$ , no discontinuity) —  $\Delta = 1$  is selected at birth, with no barrier ever crossed. The corollary is as important as the result: for long-range/mean-field systems metastability is *exact* — there is no finite nucleation droplet [47], so every horizon keeps its birth branch forever. A black hole, created abruptly at macroscopic  $N$  by collapse of a low-entanglement (vacuum-like) region, is born *in the area well* and locked there ( $N \cdot \Delta f \sim 10^{77}$  for a stellar hole, against  $\ln(t_0/t_P) \approx 140$ ):  $\Delta_{\text{BH}} = 0$  is derived, and the same  $e^{-N}$  suppression that forbids relaxation-based selection protects the black-hole prediction of §7.1. The  $10^{-7}$ -drive vs  $10^{61}$ -dof objection never arises, because nothing is ever flipped. *Flag, quantified*: the mean-field landscape is here extrapolated to  $N \sim O(1)$

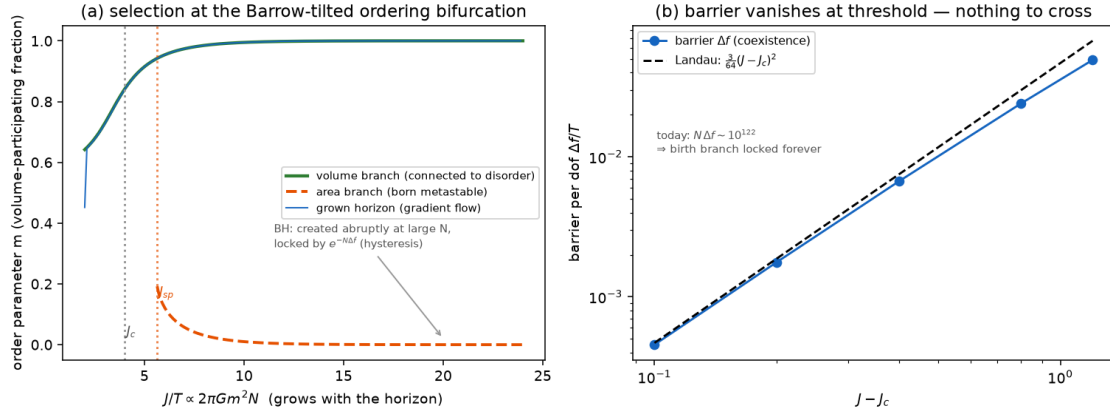


Figure 4: Selection at the ordering bifurcation. (a) With the Barrow tilt, the minimum continuously connected to the disordered state is the volume branch (green); the area branch (orange, dashed) is born *metastable* only at the spinodal  $J_{\text{sp}} > J_c$  and is never occupied by the grown horizon, whose gradient-flow trajectory (blue) tracks the volume branch with no discontinuity. A black hole, created abruptly at large  $N$ , starts on the area branch and is locked there by exact long-range hysteresis. (b) The barrier vanishes quadratically at threshold — Landau’s  $3(J - J_c)^2/64$  — so the grown horizon crossed the bifurcation when there was nothing to cross; today’s  $N \cdot \Delta f \sim 10^{122}$  locks the birth branch forever.

— the quantum-gravity era — and we stress-test rather than assume the adiabaticity. A Kibble–Zurek scan of the swept, tilted bifurcation with per-dof noise  $\sim 1/N$  finds: (a) at the *physical* Barrow tilt  $b_{\text{phys}} = \sqrt{N} - 1$ , selection is deterministic ( $P = 1.000$ ) for  $N \geq 10$  at every sweep rate; (b) the untilted control is a fair coin, confirming Barrow’s slope (i) as the selecting agent; (c) in the fully self-consistent sweep ( $J = 2\pi N$ , so the horizon is born already weakly ordered), selection is equilibration-limited —  $P(\text{volume})$  rises monotonically from 0.64 (absurdly fast) toward 1 with sweep time, reaching  $\approx 0.9$  at the slowest rates tested, and is volume-biased in *every* regime. The physical clock ( $\geq N^{3/2}$  relaxation times per e-fold of growth) sits on the slow/equilibrated side, but its  $O(1)$  normalisation at  $N \sim 2$  is genuinely Planck-era physics: the precise form of the selection claim is therefore a *probability*  $\geq 0.9$  rising to 1 in the equilibration limit, never a coin flip, and never area-biased — with the  $\Delta$  measurement (§7) testing the outcome for the one horizon we have.

- (iv) The horizon’s membrane hydrodynamics — and why no *observable*  $z^*$  transient is claimed. The apparent horizon has first-principles hydrodynamics: transverse redistribution is locally conserving (the Damour–Navier–Stokes membrane equation, with the *dynamical/apparent*-horizon bulk viscosity positive,  $\zeta = +1/16\pi G$  [48], unlike the teleological event horizon’s negative value), and two coefficients are *exact* — the sink rate, derived symbolically from the Cai–Kim definition  $\kappa = (1/2\sqrt{-h})\partial_a(\sqrt{-h} h^{ab}\partial_b \tilde{R})$  on the flat-FRW apparent horizon,  $\kappa = -H(1 - 3w_{\text{eff}})/4$  ( $H/4$  in the matter era,  $H$  in de Sitter, zero at radiation domination), and the deposition source, from the perturbed Misner–Sharp condition,  $\delta R_A/R_A = -\frac{1}{2}\delta M/M$ . One might read these coefficients as the dissipation of a *roughening* horizon surface and infer a near-conservative, self-organised-critical  $z^*$  transient — a truncated  $\tau \approx 3/2$  avalanche window and a deterministic  $\epsilon$ -lag in  $f\sigma 8$ . **We do not claim such an observable** (the case is set out in §6). The companion count paper [49] shows the classical

apparent-horizon *shape* is constraint-slaved: the  $\theta_+ = 0$  condition is elliptic ( $\nabla_{\mathcal{S}}^2 h + 2h + 2h^2 = 2\mathcal{S}$ , its eq. 2.9), with no  $(\nabla h)^2$  and no  $\partial_t h$  at any order, so the horizon has no growth equation and does not roughen — there is no height field to carry an avalanche dynamics, and the count paper’s bond-cut analysis finds no roughening universality class either. The membrane coefficients above stand as exact GR facts, but they do *not* source a critical  $z^*$  observable. What survives is the *drive* itself (next paragraph), which populates the volume capacity selected at birth; the near-critical/SOC layer and the  $\epsilon$ -lag it would carry are not claimed (§6).

The drive is a violent-relaxation deposit, and the  $\gamma$ -mechanism is the same event. Structure forms by gravitational collapse, i.e. by *violent relaxation* [50], which drives a long-range system on the dynamical time to a quasi-stationary state rather than to thermal equilibrium. This single event does two things SEDE otherwise treats separately. First, it dissipates the halo binding energy  $E_{\text{bind}} = M\sigma_v^2 \propto M^{5/3}$  (virial:  $R_{\text{vir}} \propto M^{1/3}$ ,  $\sigma_v^2 \propto M^{2/3}$ ) into the horizon ledger — the mass weight that *is* the structure coupling  $\gamma \approx 1.5$  (the cosmology paper, Prescription 4C) [5]. Second, that same deposited binding energy ( $\propto \langle \sigma_v^2 \rangle$ ) is the amplitude of the structure drive that populates the volume capacity. So the  $\gamma$ -weight and the drive are one quantity from one process — this identification is robust, and independent of the excluded near-critical layer (result iv). Two corollaries follow: collapse is prompt ( $t_{\text{dyn}}/t_{\text{Hubble}} = \Delta_{\text{vir}}^{-1/2} \approx 0.07$ ), so the horizon is driven by an ongoing sequence of such events whose envelope is  $f_{\text{sat}}$ ; and, once the volume branch is selected, the quasi-stationary state’s lifetime in a long-range system scales as  $N^\delta$  [7], so at the horizon’s  $N \sim 10^{122}$  it is permanent — a second permanence argument beside the second-law one (route E). Both the halo QSS and the de Sitter horizon have negative specific heat, consistent with the microcanonical (volume) selection of §4.

## 6 The $z^*$ transient: why none is claimed

It is natural to ask whether the structure drive produces a *distinctive* observable at  $z$ . If the drive arrived on a nearly conservative roughening\* horizon (§5 iv), the  $z^*$  response would carry mean-field self-organised-criticality statistics — a scale-free  $\tau \approx 3/2$  avalanche window, truncated at a computable cutoff  $s_c = 2/\epsilon^2$  — whose survey-accessible face would be a deterministic  $\epsilon$ -lag in  $f\sigma 8$  (a  $z^*$ -localised, growth-locked distortion,  $S/N \approx 0.5\text{--}1.5$ ). **We claim no such observable**, for a first-principles reason. The companion count paper [49] shows the classical apparent horizon is constraint-slaved and has no growth equation (its eq. 2.9:  $\theta_+ = 0$  is elliptic, with no  $(\nabla h)^2$  and no  $\partial_t h$  at any order), so it does not roughen; there is no height field, no roughening universality class, and hence no self-organised-critical avalanche dynamics of the horizon and no  $\epsilon$ -lag derived from one.

The exclusion is surgical. It does **not** touch the selection of  $\Delta = 1$ , which is fixed at the horizon’s birth (§5 iii) independently of any  $z^*$  dynamics, nor the exact membrane coefficients of §5 (iv) ( $\kappa = -H(1 - 3w_{\text{eff}})/4$ ,  $\delta R_A/R_A = -\delta M/2M$ ), which stand as GR facts. The empirical cost is minimal: such an  $\epsilon$ -lag would be marginal ( $S/N \approx 0.5\text{--}1.5$ ) and the scale-free layer is in any case *not* two-point observable, so the model loses no decisive test — the decisive prediction remains the point value  $\Delta = 1$  (§7), decided by DESI DR3 + Euclid. The frozen falsifier matrix carries no  $z^*$  two-point entry accordingly. The parent predictions carried by the cosmology paper (§6) [5] (the CHR enhancement, P3–P5) rest on the same roughening premise and receive the same treatment; that reconciliation is separate and is not taken here.

## 7 What would make it a closure

A closure derives  $F(m)$  itself, reducing the dark sector to  $\Omega_m + \textit{established}$  physics. The three routes converge:

- **A (DSSYK).** Entropy extensive in the dof count ( $S_{\max} = (N/2)\ln 2 \propto N$ , slope 1.00); cannot be super-extensive ( $\textit{volume} \propto N^{3/2}$ ). We state this plainly: read in the standard Narovlansky–Verlinde dictionary, the leading concrete model of de Sitter holography maps to the de Sitter *area* (Gibbons–Hawking) — i.e. it currently favours the area branch, a real tension SEDE’s volume postulate must eventually overcome. The mitigating point — that DSSYK is all-to-all / geometry-free and so cannot, by itself, *pose* the bulk-vs-boundary counting question — is secondary, not a neutraliser. As things stand, route A is evidence to be answered, not used.
- **B (Euclidean dS saddles).** Einstein and Barrow horizon free energies each have a single saddle — no volume-law saddle of the bare path integral (route-E no-go in saddle form).
- **C (thermal  $F(m)$ ).** The bistable landscape *is* derivable —  $F(m)/T = -(J/2)m^2 + \theta m + [m \ln m + (1-m)\ln(1-m)]$  from gravitational cooperativity ( $J = \lambda_{\max} \geq J_c$ ), the holographic pull (route E), and configurational entropy: area well  $m \approx 0.07$ , volume well  $m \approx 0.93$  whenever  $J \geq J_c$  — and each term is now sourced independently (exact counting; pairwise-gravity  $m^2$ ; mean-field exactness for  $\alpha < d$ ), so route C’s status is upgraded from “assumed form” to “derived given the two-state dof idealisation”. Level-2 existence, conditional on the volume branch being accessible.

All three reduce to one statement — *the horizon’s dof count is the bulk (volume) one* — which A and B cannot supply and C uses. Closure has three precise forms: **Level 1**, a dS-holography computation of the bulk count (open, not shortcut-able past route E; posed algebraically in §7.1b); **Level 2**, exhibiting the volume branch as a real saddle/state (C gives the landscape conditional on it); **Level 3**, *measuring*  $\Delta$  (DESI DR3 + Euclid to  $\sim 0.09$ ; the pre-registered point  $\Delta = 1$  establishes,  $\Delta = 0$  refutes, an intermediate value falsifies the locked- $\Delta$  picture — decisive, in hand; the predictions are frozen with an amendment protocol in the pre-registration).

Level 3 already has a present-data instalment. On DESI DR2 BAO + Pantheon+ + CMB distance priors + cosmic chronometers +  $f\sigma_8$ , in one CAMB-in-the-loop pipeline, the un-gated Hubble-cutoff Barrow family pays  $\Delta\chi^2 = +2416$  to sit at  $\Delta = 1$  — reproducing, in this related family, the community result that bare Barrow HDE cannot live at the volume endpoint [51]. The gated family, by contrast, pays only  $\Delta\chi^2 = +0.8$ , with a profile-likelihood interval  $\Delta = 0.93$ , 68% CL [0.83, 1.02] (all base parameters re-optimised at each  $\Delta$  — a profile, *not* the full marginalised posterior, which remains the DR3 + Euclid target; the label is precise because the cosmology paper (§5.6) [5] reserves “marginalised measurement” for that future). The pre-registered window lies inside the interval, and  $\Delta = 0$  is excluded by  $\Delta\chi^2 \approx 371$  — a *profile* statement: read as such, it overstates the marginalised model-level signal (the cosmology paper’s  $\Delta\text{DIC} \approx -3$ ) by roughly two orders of magnitude in  $\chi^2$ , and must not be quoted as a posterior exclusion. SEDE at  $\Delta = 1$ , carrying zero dark-sector parameters, fits these data 2.9  $\chi^2$ -units *better* than  $\Lambda$ CDM at equal parameter count, and within 1.4 AIC-units of  $w_0$ w $\Lambda$ CDM, which spends two extra parameters. The structure gate is, quantitatively, what makes the volume-law count observationally viable — on data already public.



## 7.1 The count is state-dependent — and that resolves the “everything points to area” tension

Routes A and B, and the standard handles generally (the de Sitter modular Hamiltonian is a *local* boost; DSSYK in the Narovlansky–Verlinde dictionary; the CKN holographic bound; black-hole thermodynamics), all favour the area count. Read as universal statements about “the” horizon count they look like a standing objection. The resolution is that they are not universal: the count is state-dependent, and every one of those handles is an *equilibrium* horizon. The equilibrium-vs-driven discriminator of §4 is exactly this — the  $f_{\text{sat}}$  gate *is* the fraction of the volume capacity that structure-driving activates,  $N_{\text{eff}} = N_{\text{area}} + (N_{\text{vol}} - N_{\text{area}}) \cdot f_{\text{sat}}$ , so the equilibrium limit ( $f_{\text{sat}} \rightarrow 0$ ) is area and the driven, locked limit ( $f_{\text{sat}} \rightarrow 1$ ) is volume. (This is *not* the rejected running- $\Delta$  background of §3: the cosmic horizon is carried through the ordering bifurcation at birth and *locks* at constant  $\Delta = 1$  (§5 iii); the state-dependence is *across horizons*, driven cosmic vs equilibrium black hole, not a running  $\Delta$  over cosmic time.) So the area results are SEDE’s equilibrium baseline, not counter-evidence. One case needs care because it is *not* obviously equilibrium: the DSSYK correspondence is a duality with the de Sitter static patch — the cosmological horizon itself. The point is that it is a duality with *eternal* (matter-free) de Sitter, an equilibrium spacetime with no structure to drive it ( $f_{\text{sat}} \equiv 0$ ); the real universe’s apparent horizon is embedded in a matter-plus-structure cosmology and is driven. So “DSSYK = area” is the area of *equilibrium* de Sitter — again the  $f_{\text{sat}} \rightarrow 0$  limit — and is consistent with a volume-law *driven* horizon, not a contradiction of it. Two derivations remove what used to be this argument’s “one layer out” hedge. (a) The gate is kinetics, not a convention. The minimal absorption law — each violent-relaxation deposit activates a fraction of the *remaining* inactive volume capacity in proportion to the deposit,  $df = \gamma(1 - f) d(D^2)$  with  $f(D \rightarrow 0) = 0$  — has the solution  $f = 1 - e^{-\gamma D^2}$ : *exactly* the cosmology paper’s gate, verified against the real growth history to  $10^{-5}$ . The interpolation  $N_{\text{eff}} = N_{\text{area}} + (N_{\text{vol}} - N_{\text{area}}) \cdot f_{\text{sat}}$  is then the *occupancy* statement of the same kinetics, not an independent assumption, and the equilibrium horizons (black hole, eternal dS:  $D \equiv 0$ ) are its  $f = 0$  limit with no extension required. The alternative reading —  $f_{\text{sat}}$  as the *equilibrium* global-minimum map  $m(\theta)$  of the §5 landscape — *is rejected by shape: deep in the ordered phase that map switches discontinuously at coexistence (a step, max slope  $\sim 300$  vs the gate’s  $\sim 2$ ), so the smooth, growth-tracking gate can only be a cumulative kinetic\* quantity* — which is the driven-NESS picture stated mechanically. Under the birth-selection of §5 (iii) the division of labour is clean: the *branch* is fixed at birth;  $f_{\text{sat}}$  is the *occupancy* of the volume capacity, not the branch selector. (b) The equilibrium-area ceiling is an algebra type. In the crossed-product analysis of Chandrasekaran, Longo, Penington & Witten [52], the *eternal* dS static patch (with observer) has a Type  $\text{II}_1$  algebra — the unique case possessing a maximum-entropy state, and that maximum *is* the Gibbons–Hawking area — while the black-hole exterior is already Type  $\text{II}_\infty$  (entropy unbounded above). “ $S \leq \text{Area}$ ” is thus the  $\text{II}_1$  ceiling of the *equilibrium* algebra, not a law of horizons. Two further results sharpen this to its final form. First, a negative result: the algebra *type* is fixed by the infrared attractor — bounded horizon entropy (any accelerating attractor,  $\Lambda$ CDM and SEDE alike, since SEDE too ends at finite  $E_\infty = \Omega_{\text{DE}}$ ) is  $\text{II}_1$ -compatible, while only eternally decelerating cosmologies force  $\text{II}_\infty$  — so the type does not discriminate the counts. Second, and decisively: in a Type II factor the trace — and hence the *value* of the maximal entropy — carries a free normalisation. CLPW fix it by calibrating to the Gibbons–Hawking entropy of the semiclassical Einstein saddle (their entropies are explicitly defined up to an additive constant). “The eternal-patch ceiling is the area” is therefore an Einstein-saddle calibration, not an algebraic output — the same  $\text{II}_1$  structure calibrated to the volume count is equally consistent, matched to a modified



gravitational sector. The residue thereby stops being an independent open question: the algebraic unknown is the trace calibration, the calibration is the count, and the count is  $\Delta$  — the Level-3 measurement answers the Level-1 question restated (a standalone treatment is given in the companion trace-calibration note). The DSSYK corollary sharpens likewise:  $\dim \mathcal{H} = 2^{N/2}$  *by construction*, so DSSYK cannot pose super-extensivity for any state; whether its  $N$  scales as  $R^2$  or  $R^3$  is a UV input to the dictionary, not an SYK output.

This makes the residue doubly falsifiable, on two independent horizons. (i) The cosmic horizon (driven) should have  $\Delta = 1$  — DESI DR3 + Euclid. (ii) The black hole (equilibrium) should have  $\Delta = 0$  — and the observable is primordial-black-hole evaporation: SEDE *predicts that PBHs evaporate normally* (standard Hawking), because black holes are undriven, whereas a *universal*-volume theory forbids it ( $T_B/T_H \sim 7 \times 10^{-21}$ , evaporation time  $\times 10^{81}$ ). The PBH channel therefore tests the *mechanism* — it distinguishes driven-NESS SEDE ( $\Delta_{BH} = 0$ ) from universal-volume quantum gravity ( $\Delta_{BH} = 1$ ) — and the black hole being area-law, far from a problem, is a SEDE prediction. This is a *future* discriminator, not present-day evidence: SEDE’s  $\Delta_{BH} = 0$  means standard Hawking evaporation, consistent with all current data (including  $\sim 10^{15}$  g PBH abundance bounds) precisely because it predicts no new black-hole effect; the channel discriminates only on a positive PBH-evaporation detection or a robust population statement.

This also yields a new statement about black holes in general: area-law is the *equilibrium* default, not a fundamental geometric law. A black hole is area-law *because of how it is born*: created abruptly at macroscopic  $N$  from a low-entanglement (vacuum-like) region, it starts in the area well, and exact long-range hysteresis (§5 iii) locks it there with Kramers exponent  $N \cdot \Delta f \sim 10^{77}$ . It is also undriven ( $f_{\text{sat}} \rightarrow 0$ ), so neither mechanism ever populates its volume capacity. The cosmic horizon’s uniqueness is a calculation, not an assumption: it is the only horizon *grown* continuously from Planckian size through the ordering bifurcation. The most-driven black hole conceivable is a primordial one *at formation*, where the entire horizon is built by a collapsing overdensity (violent relaxation); yet even there the structure entropy deposited is negligible against the black-hole entropy *created*,  $S_{\text{rad}}/S_{BH} \sim (M/M_P)^{-1/2}$ , so the formation drive  $f_{\text{form}} \sim (M_P/M)^{1/2} \ll h_c$ , the gate-activation threshold, by many orders. Accreting and merging black holes are weaker still. So SEDE robustly predicts *all* black holes are area-law ( $\Delta_{BH} = 0$ ), and the cosmic apparent horizon — uniquely driven *sustainedly* to  $f_{\text{sat}} \rightarrow 1$  over the whole structure-formation history — is the only volume-law horizon. The falsifier is correspondingly sharp: an *isolated* volume-law black hole (anomalously slow PBH evaporation, or a modified-area GW ringdown) would mean  $\Delta$  is *universal* (Barrow-type) and would refute the state-dependent, driven-vs-equilibrium mechanism — so SEDE is *more* falsifiable on black holes than a universal-Barrow theory, not less.

The theoretical residue is correspondingly narrowed: not “derive the count,” but two named statements — “the branch is set at the birth bifurcation” (§5 iii, derived within the  $F(m)$  landscape,  $N \sim 1$  extrapolation flagged) and “the deposits populate the volume capacity” (the gate kinetics, §7.1a) — both now *consistent with* the equilibrium area results (their  $f_{\text{sat}} \rightarrow 0$  / abrupt-birth limit), with the two-channel measurement deciding it either way regardless of dS holography.

Two final remarks, stated openly. First, the *magnitude* worry — a  $10^{-7}$  structure drive against a  $10^{61}$  change in the activated count — is dissolved rather than answered: under §5 (iii) the drive never flips anything. The branch (the  $10^{61}$  capacity) is fixed at birth by the bifurcation; the  $10^{-7}$ -per-Hubble- time deposits merely *populate* it, cumulatively, through the derived gate kinetics (§7.1a), and the two numbers no longer need to be compared. What remains modelled is the landscape’s two-state idealisation (§5 ii), not a heavy-lifting amplification. Second, the *pre-registered* case: the prediction is a *point*,  $\Delta = 1$ , fixed by

the birth-selection of §5 (iii) — not a window read off a roughening exponent. A measured cosmic  $\Delta$  significantly below it — say  $\Delta \approx 0.5$  from DESI DR3 + Euclid,  $\sim 5\sigma$  from  $\Delta = 1$  at  $\sigma_\Delta \approx 0.09$  — would falsify the locked- $\Delta = 1$  picture outright and would *not* be a free parameter to absorb.

**The count as a horizon Hausdorff dimension — and why the roughening reading is superseded.** The Level-1 question — is  $\dim \mathcal{H}(\text{de Sitter static patch}) = e^{\text{Area}/4}$  or  $e^{\text{Volume}}$ ? — has a clean geometric form:  $\dim \mathcal{H} = e^{\text{Area}} \iff$  the horizon is a smooth 2-surface (Hausdorff dimension  $d_H = 2$ ), and  $\dim \mathcal{H} = e^{\text{Volume}} \iff$  a space-filling fractal ( $d_H = 3$ ), so the count *is* the horizon’s Hausdorff dimension,  $\Delta = d_H - 2$  — a reframing of “area or volume?” as an empirically-decidable *geometric* statement. One could try to turn that into a *dynamical* ceiling by treating the driven horizon as a roughening surface (a roughening 2-surface has  $d_H \in [2, 3]$ , so  $\Delta \leq 1$ ). That roughening reading is **superseded** by the companion count paper [49], which does the same job more strongly and without the surface-height picture that §6 sets aside: (i) the ceiling  $\Delta \leq 1$  is a *theorem* from a finite per-site leg budget (bond-cut capacity  $C \leq \kappa \cdot |\Omega|$ ), not a fractal argument — matching the bound the cosmology paper (§8.1) [5] claims and the dof-counting reading of §8.5; (ii) the count is *discrete*,  $\Delta \in 0, 1$  — the count paper finds no minimum-cut surface for a long-range network, hence no height field and no roughening universality class, so  **$\Delta$  is a count, not a Hausdorff dimension** (exactly the cosmology paper’s §8.5, and confirmed classically by the constraint-slaved apparent horizon, eq. 2.9); and (iii)  $\Delta = 1$  is the infinite-cutoff ( $L \rightarrow \infty$ ) endpoint for the flat, infinite spatial slice — a stated cutoff choice in that paper, not a derivation of the value. The equilibrium (undriven) horizon is the area endpoint, consistent with the area frameworks (Gibbons–Hawking, the Type II<sub>1</sub> maximum [52], DSSYK) as descriptions of the *eternal* horizon. Within the present paper the *value*  $\Delta = 1$  is set by §5 (iii)’s birth-selection, which lands the grown cosmic horizon on the volume branch as a point; the geometric reframing ( $\Delta = d_H - 2$ ) and the ceiling  $\Delta \leq 1$  are what the count question reduces to, now carried out in the count paper (the ceiling as a model-conditional theorem; the value  $\Delta = 1$  as the stated  $L \rightarrow \infty$  endpoint choice). We do not resolve de Sitter holography here: the sharpest in-paper statement of what a resolution requires remains the II<sub>1</sub> trace calibration of the FRW patch, which §7.1b shows *is* the  $\Delta$  measurement asked algebraically. The empirical  $\Delta$  (DESI DR3 + Euclid, §7) is the arbiter.

## 8 Discussion

The chain reformulates SEDE’s foundational input — and, in its present form, earns back a *qualified partial reduction*. The residual list has contracted to: the tensor-network description of the horizon state (the  $S_{\text{ent}} = S_{\text{grav}}$  tier restated micro-structurally, §5 ii — the two-state dof is now the derived Ising dual of that description); the Planck-era clock normalisation behind the birth-selection probability (quantified, not assumed:  $P(\text{volume}) \geq 0.9$ , rising to 1 in the equilibration limit, volume-biased in every regime); and one scoped coefficient (the transverse diffusivity of the height equation, whose sink rate and source are now exact GR results,  $\kappa = -H(1 - 3w)/4$  and  $\delta R_A/R_A = -\delta M/2M$ ). The former “open dS-holography count” is no longer on the list as an independent item: the algebra type is fixed by the infrared attractor, and the ceiling’s value is a trace calibration — the count *is* the  $\Delta$  measurement, asked algebraically (§7.1b; and the standalone trace-calibration note). Assuming the tensor-network tier, the selection is a single causal chain from one premise, gravity’s long-rangedness: exact mean-field thermodynamics sources the two-phase F(m) term-by-term (§5 ii); the N-dependent coupling  $J/J_c = 2\pi G m^2 N$  forces the ordering bi-

furcation at Planckian size, where Barrow’s monotone slope (§5 i) tilts the nascent well — so the grown cosmic horizon tracks onto the volume branch,  $\Delta = 1$  selected at birth with probability  $\geq 0.9$  ( $\rightarrow 1$  in the equilibration limit), with no barrier ever crossed (§5 iii); and exact long-range hysteresis locks every horizon to its birth branch, making the black hole’s  $\Delta_{\text{BH}} = 0$  a derived prediction rather than an accommodation. No  $z^*$  near-critical transient is claimed (§6): the companion count paper [49] shows the apparent horizon is constraint-slaved (its eq. 2.9) and does not roughen, so the SOC/ $\epsilon$ -lag layer that §5 (iv) might suggest does not survive — the exact membrane coefficients of §5 (iv) remain as GR facts, but they source no critical observable (§6). The genuinely established *reductions* in the chain are thermodynamic: ensemble inequivalence for the long-range de Sitter horizon ( $C = -2S < 0$ ) removes the canonical area-law branch (§4), and the gate  $f_{\text{sat}}$  is derived from one-line activation kinetics (§7.1a). The reformulation yields falsifiability where the postulate had none:  $\Delta$  is pinned to the pre-registered point  $\Delta = 1$  (§5 iii; the ceiling  $\Delta \leq 1$  is a theorem in the companion count paper [49], §7.1; frozen with an amendment protocol in the pre-registration), measured to  $\sim 0.09$  by DESI DR3 + Euclid (shared with the cosmology paper), with any intermediate value falsifying — the count paper makes the count discrete ( $\Delta \in 0, 1$ ), so there is no intermediate value to accommodate. And the mechanism is already tested on public data: the gate rescues  $\Delta = 1$  by  $\Delta\chi^2 \approx 2400$  on DESI DR2, with the gated family preferring  $\Delta = 0.93$  [0.83, 1.02] at zero dark-sector parameters (a profile likelihood, §7, Level 3). The residue is a clean hand-off — now in its sharpest form a *measurement*, not an open theory question — and the cosmology stands independently (the cosmology paper).

## 8.1 The covariant completion and the amplitude (status update)

Two clarifications supersede earlier framings of this programme and should be read as governing. **(i) The completion is local, and the parity argument is retracted.** The claim that an  $H$ -linear density  $\rho_X \propto H$  *cannot* descend from a local conservative action is **false**: Kinetic Gravity Braiding (Deffayet–Pujolàs–Sawicki–Vikman 2010, arXiv:1008.0048) is an explicit local counterexample, and the completed theory is a finite cubic-KGB action  $\mathcal{L} = M_{\text{P}}^2 R/2 + K(X, \phi) - G_3(X, \phi) \square \phi$  (FPAB-SEDE) whose linear perturbations are ghost/gradient-stable with  $c_T = 1$  (the cosmology paper §3.4). The former time-reversal “parity no-go”, the unique- $\Delta = 1$  *selection* claim, and the reading of  $\rho_{\text{DE}} = T s$  as a dark-sector *identity* are accordingly withdrawn:  $\rho_X \propto H$  and  $\Delta = 1$  are the phenomenological branch being completed, not consequences of a parity theorem, and the birth-selection of §5 is a *motivated conditional*, not a derivation (§8 above). Israel–Stewart bulk viscosity does **not** establish perturbative stability across the crossing (the single-fluid principal symbol diverges at  $\rho + p = 0$ ); braiding does. **(ii) The amplitude is not derived.** The scale  $\mu \approx 28.5$  MeV is the coincidence problem in dimension-three form, shared with  $\Lambda$ ; the counting/network construction motivates the *form*  $\Delta = 1$  but does not fix  $\mu$ . A numerical relation  $\mu^3 \simeq \alpha_{B0} f_\pi^2 m_\pi^2 / m_{\eta'}$  reproduces the closure value to 1.01% (predicting the braiding fraction  $b \approx 0.206$ ); we present this as a **coefficient-selection hypothesis**, not a Standard-Model derivation — the topological-coefficient identification is an ansatz and the structure gate is not obtained from QCD. The genuine microscopic home of the gate is the 2PI/closed-time-path effective action of a structure reservoir (reciprocal matter–scalar stress exchange constrained by the Bianchi identity); its kernel is not derived and it is labelled an extension, not the established core. **(iii) The dissipation is driven, not thermal — an FDT/KMS retraction.** An earlier companion derivation presented the dissipative dark-energy dynamics as a dynamical-KMS structure at the horizon temperature, with  $\zeta > 0$  identified with the fluctuation–dissipation lock  $\langle \xi \xi \rangle = 2\zeta T_{\text{AH}}$ . That claim is retracted, because the calculation was done: the noise that

actually drives the gate is cold and structure-sourced — set by the primordial amplitude  $A_s$ , not by the  $10^{-33}$  eV horizon temperature — and exceeds the KMS-thermal value by  $\sim 10^{57}$ . The *mean* horizon thermodynamics of the background ( $T_{AH} = H/2\pi$ , the  $H$ -linear form) is untouched, but the correct framing of the dissipative sector is the driven, entropy-producing NESS used throughout this paper, with  $\zeta > 0$  surviving as the second law — not a thermal-KMS effective field theory. We record the correction because it is itself informative: it unifies the dawn and dusk sectors of the wider programme as non-equilibrium at both ends. **(iv) The tensor sector is exactly GR here too** —  $\delta(z) = 0$ ,  $\Xi_0 = 1$ . The driven-NESS framing reproduces the cosmology paper’s tensor result, so the standard-siren prediction  $\Xi_0 = 1$  is a joint statement of both papers. Four channels close it. *(a) No new bulk fields*: the dynamical content — the horizon order parameter, the hysteresis lock, the structure reservoir — lives in the horizon *accounting* and the scalar/entropy sector; the covariant bulk completion is FPAB, whose tensor sector is exactly GR ( $\alpha_M = \alpha_T = 0$ , cosmology paper §6). The conclusion depends only on *where* the new physics lives, so it survives the open items in the §5 mechanism. *(b) Dissipative bulk channel, bounded to irrelevance*: the most uncharitable reading endows the driven medium with an effective shear viscosity damping GWs at the Hawking rate  $\Gamma = 16\pi G\eta/c^2$ ; an  $\mathcal{O}(1)$  propagation effect over a Hubble time needs  $\eta \approx 6 \times 10^7 \text{ Pa} \cdot s$ , whereas a deliberately generous kinetic estimate of the cosmic medium ( $\rho_m \sigma_v \times 1 \text{ Mpc}$ ) gives  $\approx 24 \text{ Pa} \cdot s$ , so  $|\Xi_0 - 1| \lesssim 4 \times 10^{-7}$  — five orders below the forecast siren sensitivity  $\sigma(\Xi_0) \sim 10^{-2}$ , and itself conservative (LIGO-band wavelengths are far below any structure scale, so the fluid regime does not even apply). *(c) Horizon-boundary absorption, closed by our own result*: the apparent-horizon condition  $\theta_+ = 0$  is elliptic (constraint-slaved, no growth equation; count paper eq. 2.9) — a boundary with no dynamical modes cannot resonantly absorb the sub-horizon tensor radiation ( $k/\mathcal{H} \sim 10^{18-20}$  in the LIGO band). *(d) Fluctuation–dissipation consistency*: zero tensor dissipation pairs with zero horizon-sourced tensor noise ( $\Omega_{\text{gw}} \sim (H/M_{\text{P}})^2 \sim 10^{-122}$ ). This is consistent with clarification (iii): the *scalar* gate is a cold, structure-driven NESS, while the *tensor* sector sees an equilibrium GR vacuum channel in both formulations. One conditional remains, at low risk: if the reservoir kernel is ever realised as a graviton-mediated transport channel it would add an utterly negligible *source* to the stochastic background, not a propagation modification, so  $\Xi_0 = 1$  survives even that branch.

For honesty the branch selection itself is **gravitationally silent**: the ordering bifurcation of §5 is a horizon-ledger transition with no bulk latent heat, no bubble nucleation and no propagating horizon modes, so it emits no gravitational-wave signature — its only observable trace is the  $\Delta = 1$  it leaves behind. Relatedly,  $\Delta_{\text{BH}} = 0$  (black holes locked to the area law, §5, §8) means SEDE predicts exactly standard Bekenstein–Hawking black-hole thermodynamics: unmodified Kerr quasinormal spectra and no echoes. This separates SEDE from quantum-corrected-entropy models that predict near-horizon structure — a confirmed ringdown anomaly attributable to modified black-hole entropy would falsify SEDE, and the continuing null echo searches are (weak) SEDE-consistent evidence.

## 8.2 Reproducibility

Every quantitative claim in this paper is reproduced by the accompanying code from a single entry point, each stage carrying validation assertions; all stages pass. The four figures are regenerated from the same code.

## 9 Conclusion

We set out to test how much of SEDE’s single foundational input — the volume-law (Barrow  $\Delta = 1$ ) horizon entropy — is irreducible. Decomposed into *state*, *form*, *scale*, and *count*, the first three reduce to first principles given the standard entanglement identification of the horizon entropy, and a Bekenstein no-go shows the residual count cannot follow from any energy bound. What replaces the postulate is a *selection*: grown continuously from Planckian size, the cosmic horizon passes through the ordering bifurcation of a driven long-range system with Barrow entropy tilting the nascent well toward volume, so  $\Delta = 1$  is selected at birth — with probability  $\geq 0.9$ , rising to unity in the equilibration limit — and long-range hysteresis locks each horizon to its birth branch: the cosmic horizon to volume, an abruptly-formed black hole to area.

The significance is a change of epistemic status rather than a claimed proof. An untestable counting postulate becomes a falsifiable prediction:  $\Delta$  is pinned to the pre-registered point  $\Delta = 1$  (§5 iii; the ceiling  $\Delta \leq 1$  is a theorem in the companion count paper [49]), decided to  $\sigma(\Delta) \approx 0.09$  by DESI DR3 + Euclid, with any intermediate value falsifying — the count paper makes the count discrete ( $\Delta \in 0, 1$ ). On present public data the mechanism is already evident — the structure-gated family prefers the volume endpoint at zero dark-sector parameters, which the un-gated Barrow family pays  $\Delta\chi^2 \approx 2400$  to reach. The residue is thus handed off as a *measurement*, not an open theory question. The one premise still to be earned is the tensor-network description of the horizon state, on which the selection chain is conditional; future work should target that description directly. The cosmological results of the cosmology paper stand independently of the reduction attempted here.

## Data availability statement

All code and analysis pipelines are publicly available at <https://github.com/spsingularity/sede-foundations>. Every quantitative claim in this paper is produced by the accompanying code wired into a single entry point, each stage carrying validation assertions; the figures are regenerated from the same code. The model and its cosmology live with the companion cosmology paper [5]; no additional observational data beyond the cosmology paper’s standard public inputs is used here. A tagged release is archived at Zenodo, DOI 10.5281/zenodo.21525521.

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